

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

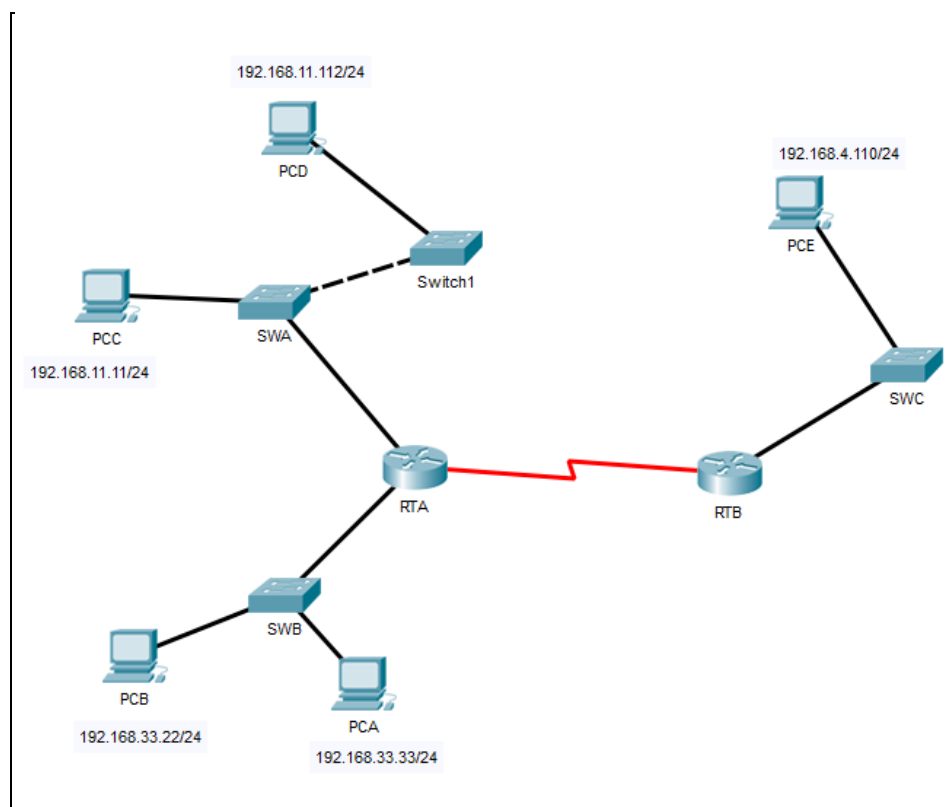


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

Switch SWA	Switch SWB	Switch SWC
<pre>SWA>enable SWA#show arp SWA#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 0002.4a00.0e91 DYNAMIC Fa0/1 1 000c.8546.7d85 DYNAMIC Fa1/1 SWA#</pre>	<pre>SWB>enable SWB#show arp SWB#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 000c.cf0c.593a DYNAMIC Fa0/1 SWB#</pre>	<pre>SWC>enable SWC#show arp SWC#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 0001.977a.b614 DYNAMIC Fa0/1 SWC#</pre>

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PC A	PC B
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>arp -a No ARP Entries Found C:\></pre>	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>arp -a No ARP Entries Found C:\></pre>
PC C	PC D
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>arp -a No ARP Entries Found C:\></pre>	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>arp -a No ARP Entries Found C:\></pre>
PC E	
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>arp -a No ARP Entries Found C:\> </pre>	

- e. What are your thoughts on the results?

There is no entry for all PC because all PC did not receive any ARP request. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.
- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here
- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PC A	PC B
<pre> C:\>ping 192.168.33.22 Pinging 192.168.33.22 with 32 bytes of data: Reply from 192.168.33.22: bytes=32 time<1ms TTL=128 Reply from 192.168.33.22: bytes=32 time=1ms TTL=128 Reply from 192.168.33.22: bytes=32 time=1ms TTL=128 Reply from 192.168.33.22: bytes=32 time<1ms TTL=128 Ping statistics for 192.168.33.22: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 1ms, Average = 0ms C:\>arp -a Internet Address Physical Address Type 192.168.33.22 0060.47ea.a746 dynamic </pre>	<pre> C:\>arp -a Internet Address Physical Address Type 192.168.33.33 0002.1755.9a06 dynamic </pre>
PC C	PC D
<pre> C:\>arp -a No ARP Entries Found C:\> </pre>	<pre> C:\>arp -a No ARP Entries Found C:\> </pre>
PC E	
<pre> C:\>arp -a No ARP Entries Found C:\> </pre>	

Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
```

- Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=13ms TTL=126
Reply from 192.168.4.110: bytes=32 time=9ms TTL=126
Reply from 192.168.4.110: bytes=32 time=10ms TTL=126
Reply from 192.168.4.110: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 13ms, Average = 10ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are in the ARP table only after PCC ping to PCB, PCD and PCE. No ARP entry before the ping process. The ARP table shows the entries when PCB, PCD and PCE receive the ARP request from PCC.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 192.168.11.1    -         0002.4A00.0E91 ARPA   FastEthernet1/0
Internet 192.168.11.11  4         00D0.D39A.C0D9 ARPA   FastEthernet1/0
Internet 192.168.33.1   -         000C.CF0C.593A ARPA   FastEthernet0/0
Internet 192.168.33.22  4         0060.47EA.A746 ARPA   FastEthernet0/0
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 5 0060.702D.7C08 ARPA FastEthernet0/0
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.4a00.0e91   DYNAMIC   Fa0/1
1       000c.8546.7d85   DYNAMIC   Fa1/1
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
```


- d. Do switches use arp table? (Y/N) **Y**
- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches use ARP table to store ARP address and MAC address for network devices. The table is use to identify the destination MAC address for communication with network nodes.

- f. What information does the command **show mac-address-table** gives?

The MAC address table entries on a switch. The 'DYNAMIC' displays the MAC address that learnt by the switch during its operation.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- a. Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network diagram shows a switch (S1) connected to a router (R1) and a wireless router (WRS2). PC1 is connected to S1 (VLAN 10), PC2 to S1 (VLAN 20), and PC3 to WRS2. A 'Subinterfaces' list for R1 is shown: G0/0.10 (172.17.10.1/24), G0/0.20 (172.17.20.1/24), and G0/0.88 (172.17.88.1/24). On the right, a window titled 'Packet Tracer – Configuring Wireless LAN Access' is open. It contains an 'Addressing Table' and a progress bar.

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the window, the progress bar shows 'Completion: 100/100' and '1/1'.

- c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

